Securing SCADA System from DDoS Attack

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Abstract: Emergency response systems, water treatment facilities, wastewater collection systems, Oil and gas pipelines, electrical power transmission systems, wind farms, defence networks, and large-scale communication networks are all vulnerable to the increasing prevalence of this threat. of distributed denial of service (DDoS) attacks. A dynamic proxy defence method is required because static thresholds cannot protect a critical SCADA system from DDoS attacks. This proposed way to protect these services from attacks is by masking their IP addresses hidden behind the proxies. This technique allows for the swift removal of misbehaving or attacked proxies by having the service generate hidden paths on the fly. By removing the attacker's cover, this technique creates a secure network that is only accessible to authorized users. An attacker doesn't need to know the data proxy to disrupt ongoing client communications because it is hidden. This DDOS prevention architecture is a useful addition to existing DDoS defences, and it can be effectively implemented in SCADA systems, data servers, and data centres. Even when they only occupy 10% of the network, a dynamic proxy density of 0.5% is sufficient to suppress over 93% of attackers.

Keywords: SCADA, DDoS, Protection, cyber-attack.

1. Introduction

Distributed Denial of Service (DDoS) attacks must be stopped from reaching SCADA systems so that critical infrastructure can keep running smoothly. Cybercriminals often go after industrial processes that are controlled and monitored by SCADA systems. The point of a distributed denial-of-service (DDoS) attack is to flood a service with too many attacks at the same time [1]. The objective of a distributed denial of service attack is to stop the target from responding to valid customer requests by sending too many packets to the target [2].

common tactic is sending corrupted packets that confuse applications or protocols, causing the victim's machine to crash or restart. DDoS attacks don't rely on a specific network protocol or system weakness, but rather on the Internet and the victim's lack of resources. The attacker uses the accessibility of the Internet to identify targets and spread malicious software. The hacker then deploys an attack infrastructure, communicating with masters who give orders to agents [3].

In 2020 [4,] Because the Internet was used for so many important things, it was a simple target for Distributed Denial of Service attacks. The projected number of DDoS attacks in 2023 is expected to exceed 15 million, which is an increase from the 7.9 million attacks recorded in 2018. [7].

2. Review of the Previous Work

The nature of the threats we face is changing, and hackers are getting more advanced. Protecting critical infrastructure from DDoS attacks requires staying abreast of the latest security developments and actively monitoring SCADA systems for potential threats [11][12].

SCADA systems are essential to the operation of many businesses. There is a growing risk of cyberattacks, especially distributed denial of service attacks. The objective of a distributed denial of service (DDoS) attack is to inundate a target with an excessive amount of network traffic, causing it to become inaccessible to its intended users. DDoS attacks present a substantial threat to SCADA infrastructure [13]. However, many safeguards can be set up to reduce the possibility of an assault. Given the potentially catastrophic effects of a successful DDoS attack on SCADA systems, securing these systems is of the utmost importance. To prevent economic, environmental, and public safety disasters from disrupting critical infrastructure and industrial processes, SCADA system security must be a top priority [14][15].

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2.1 Problem

Different types of attack sources can be set up and different packet-transmission strategies and packet forms can be used in DDoS attacks to disrupt the victim's service availability. To reduce the impact of DDoS attacks, defence systems must employ a wide range of DDoS detection techniques. Depending on the computational processes involved, this may be too slow to be of real-time use for making enhancements. Coverage of security before, during, and after an incident is presented as part of the DoS management framework. To do this, we give a complete plan and specific suggestions for how to make the technical and operational parts of DoS-attack protection is a priority.

- The ability to identify attacks as they happen.
- Taking the right measures in response to both ongoing and potential threats.

Statement for this Proposed Work

This research proposal aims to develop a solution for securing SCADA systems from DDoS attacks. In addition, the proposed solution will be made scalable so that it can be used in SCADA networks.

2.2 Methodology

The method employs hidden paths that are generated dynamically and are entirely under the service's command, allowing for swift elimination of any proxy that is misbehaving or under attack. Because of this restriction, the proxy infrastructure itself may be attacked instead of the service itself. Separating the proxies into setup and data proxies is one way to counteract this kind of attack. Data proxies grant continued access to the service, while setup proxies only allow the client to make the initial connection. However, IP anycast is used by the setup proxies to partition the network into separate zones. Requests for connections made from within the geographic boundaries of an anycast setup proxy will be forwarded to that proxy by default. To secure SCADA, the focus has largely been on authentications using cryptographic techniques, which disperses the attacker's hat. On the flip side, there was some enthusiasm for strengthening security measures, bolstering system management tools, developing robust systems, and toughening up systems and services.

Implement a layered security approach. This means using multiple security measures, such as proxy infrastructure, authentication, authorization, encryption, and network segmentation, to protect your SCADA systems.

3.0 Contributions

This paper presents a proposed architecture for hiding the location of the destination from direct connections and intermediaries that do not have access to the destination's IP address. This architecture protects against distributed denial of service attacks (DDoS) with the help of anycast SPs. Because the architecture disperses and dilutes potential attackers, it is safer and simpler to find proxies. When using anycast SPs, attackers can be contained within smaller regions, making it simpler to track down or solve client puzzles. Using this model, the service can provide data proxies to its legitimate users that are commensurate with their trustworthiness, preventing disruptions to the service. Using the multicast routing capabilities of already-existing routers, it is possible to conceal destination networks entirely, build hidden paths, and quickly add or remove such paths as needed.

3.1 Challenges in Location Hiding

In location-hiding architectures, the destination must remain concealed even in the event that proxies are compromised or exhibit malicious behaviour. It is essential to consider reducing the number of trusted components, controlling the information disclosed to the proxies, and regulating the path taken by the proxies to reach the service's location. If an attacker already has the target's address, they can bypass the proxy and go straight for their intended victim. A protective filtering perimeter around the destination can be set up to ward off such threats, reducing the amount of malicious traffic that can reach the target even if the destination address is compromised. Filtering, however, presents its unique challenges, such as the need for a sufficiently large filtering perimeter to account for bottlenecks close to the destination network.

The service must be robust against attacks on the proxies for location-hiding architectures to be effective. The attacker will now focus on the proxies rather than the original target. Using a large number of proxies and switching them out regularly is one way to deal with attacks on the proxies. However, the proxies must be lightweight so that large numbers can be recruited and deployed rapidly; this includes not requiring any specialized hardware or software and not storing any private user data. One way to acquire a sizable number of proxies is to recruit them from end hosts, such as those found in P2P networks.

Proxies must be assigned and managed by location-hiding architectures in a way that minimizes the risk to legitimate users. While expanding the number of proxies can make the target more diffuse, an attack on a single proxy could still affect a legitimate client. The attacker may aim for particular proxies to disable or otherwise disrupt specific parts of the proxy network. When facing massive DDoS attacks,

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the destination must allocate and manage proxies in a way that protects legitimate users as much as possible.

Finally, clients must have a scalable and reliable discovery mechanism to learn about the proxies in location-hiding architectures. Clients can use the discovery mechanism to locate the proxies they need to access the service. This discovery service, though, needs to be resilient against attacks and able to deal with frequent shifts in the proxy set, or it will quickly become useless a single point of failure for the entire system.

3.2 DDoS Attack and Flash Crowd

A distributed denial of service (DDoS) attack or a flash crowd can lead to an overwhelming influx of visitors to a website or service. In contrast to a deliberate attempt to overwhelm and disrupt a target's online services, which is what a distributed denial of service (DDoS) attack is, a flash crowd is an unscheduled and temporary increase in legitimate user traffic caused by the interest of real users [9][10]. In terms of what drives them, what they're used for, how long they last, and what precautions should be taken ahead of time, these two events couldn't be more different. However, DDoS attacks require specialized defense mechanisms to filter out malicious traffic, so the infrastructure must be temporarily scaled up to accommodate genuine user interest during a flash crowd. Our work is inspired by the distinction between the way traffic volume behaves in flash crowds and DDoS attacks [5][6]. Therefore, the similarities between flash crowd flow and DDoS attacks are much greater. Similarities between DDoS attacks and flash mobs necessitate careful differentiation between the two to avoid false alarms. Distinguishing between DDoS flooding attacks and flash events is a major challenge for defenders. If we are unable to achieve this, it will have devastating effects. On the one hand, adversaries can fool our detectors by emulating flash crowd traffic patterns. However, our sensors might incorrectly flag spontaneous flash mobs as malicious DDoS attack [8].

4.0 LOCATION HIDING ARCHITECTURES

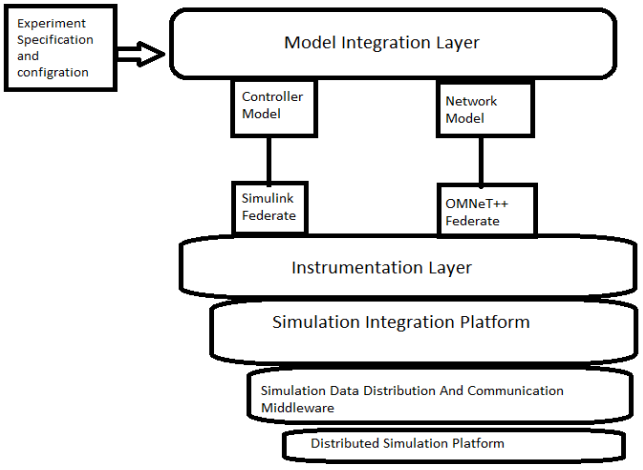


Figure 1: Proposed Simulation Architecture

Figure 1 is proposed simulation architecture of the Location-hiding scheme, such as Architecture and The Onion Router, hide the final destination's IP address and restrict access to the service to intermediaries. These architectures employ a large number of proxies to give the impression of greater reach and frequently rotate between different configurations of active proxies to confuse would-be attackers. In contrast to the Onion Router, which offers location-hidden services via an overlay-based anonymization network, the architecture in question relies on the underlying network to conceal the service and is not designed to provide anonymity. Overlay nodes, also known as "introduction points," allow a hidden destination on the Onion Router to forge secure connections to other destinations. The Onion Router's direct service can be dialed by any service-requestor to obtain an entry point list. Then, it picks one of the entry points and connects to the final hop via a third node, a rendezvous points in The Onion Router. No one knows the other person's IP address in this situation. The Onion Router's initial node and subsequent rendezvous nodes can be seen as metaphors for the architecture's settings and information, respectively. Architecture, on the other hand, is impervious to this type of attack because its backdoors always start at the network's perimeter routers. As an extra precaution, the Valet Service recommends enclosing the introduction points in a labyrinth of valet nodes. In the beginning, clients would connect to a centralized direct The Onion Router service to learn about available proxies. A Distributed Denial of Service (DDoS) attack can be launched against the Onion Router service. The architecture relies on Domain Name System (DNS) to locate available proxies, but it protects the discovery service by using standard measures to prevent DNS attacks. By creating an overlay of proxies and designating specific proxies to store

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various pieces of information about the overlay and the destination, Mayday is essentially a more generalized version of Secure Overlay Services (SOS). Anyone can access a proxy, but only authenticated users will have their traffic directed through that proxy. To protect the target, SOS sets up a buffer zone that only lets data packets from the concealed servlets through.

location-hiding architectures like Architecture and SOS face unusual difficulties when it comes to disseminating authentication information to all proxies. Because of their compact size and lack of dependence on individual user credentials, architecture proxies facilitate deployment. To protect SP customers from DP-level attacks, they employ a network of proxies. Fewer innocent users will be harmed by an attack on the service because the attacker cannot determine which DP the client is using. If a proxy is compromised by an attack, clients using SOS can easily switch to another one. Due to proxy failures, Architecture clients may experience intermittent service disruptions. Internet Indirection Infrastructure (i3) is a new system that modifies the way IP addresses are associated with real-world locations by constructing an overlay on top of the existing Internet. The client finds the service's unique identifier and sends a request to that address. The request is sent through the underlying i3 overlay to its final destination, which may provide the requesting client with a new set of private identifiers. The difficulties of penetration attacks, reconfiguration, proxy depth, and host diversity are all faced by overlay-based location-hiding techniques like SOS, i3, and The Onion Router. Due to its reliance on the routers of the underlying network, the security of the architecture can only be as good as the routers themselves. Network authenticated path panning (SNAPP) keeps senders anonymous while disregarding routing protocol decisions. Proxy servers are a good way to protect SCADA systems from DDoS attacks. But it's important to use a trusted proxy server provider, set up the proxy server correctly, and keep an eye on it for any strange activity. By following these rules, organizations can help make sure that DDoS attacks don't hurt their SCADA systems.

4.1 Identifying Misbehaving Proxies

If the proxy tries to overwhelm the destination with these fake report messages, D will be aware of it and disconnect the proxy. D, for instance, might decide on a maximum allowable path length. If D receives more than this threshold number of report messages, it will disconnect from the proxy. That is, if multiple routers have identical prefixes (for example, a /16 and a /24), the first router with the largest matching prefix is chosen as the proxy.

5.0 Result & Analysis

Using simulations on OMNeT++ we demonstrate the efficacy of anycast SPs. Based on our research, we know that an SP density of 0.5% is enough to suppress over 93% of the attackers even when 10% of the entire network is occupied by malicious users. This is true even when the density of legitimate clients is only 0.01%. Additionally, at that density, attackers are confined to a region no more than three to four hops from the SPs, and in most cases, only a single attacker is interfering with legitimate clients at a given SP.

SCADA system security against DDoS attacks can be greatly improved with the help of dynamic location-hiding architectures. Protect SCADA systems by constantly switching their IP addresses, making them more difficult to hack.

We prove the following by constructing a miniature test of architectural design:

1) By turning off route announcements for the destination network into the global Internet, the destination network can be kept completely hidden.

2) Present-day routers' support for multicast routing enables the creation of covert routes. For this goal, we can employ Single Source Multicast.

3) It only takes a few seconds, tops, to add or remove a hidden path. As a result, it would be simple to add new proxies and swiftly remove compromised proxies once they are discovered.

4) Dynamic IP address changes in enhancing the security of SCADA systems against DDoS attacks.

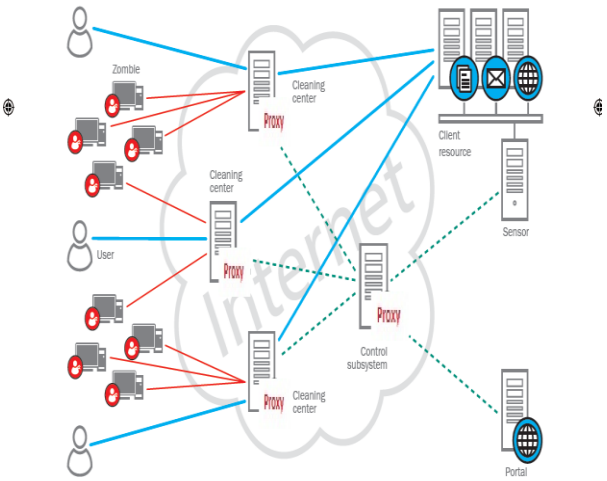


Figure 2: Proposed Defending Architecture

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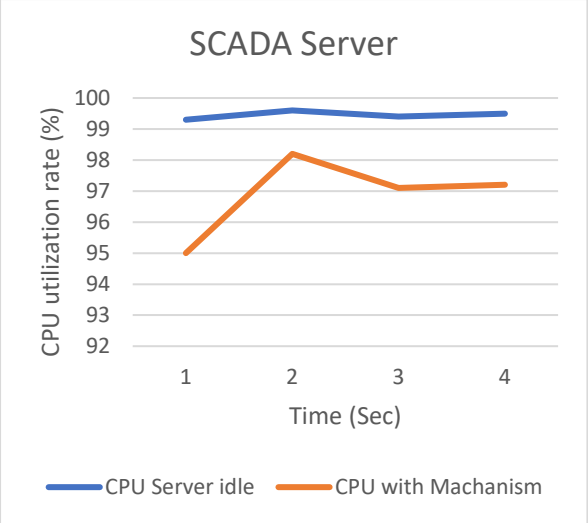


Fig 3: SCADA Server CPU utilization rate (%)

In Figure 3, The CPU of the SCADA server is observed to be in an idle state. The proportion of the SCADA server's CPU that is idle is plotted against time, which is represented on the X-axis.

Figure 2: Proposed Defending Architecture, of the defence system where dynamic proxies are there.

To lessen the impact of distributed denial of service (DDoS) attacks, dynamic proxies can be used to spread incoming traffic across several computers. As a result, the attacker's ability to disrupt the service being targeted may be hampered.

Some key points of dynamic proxies:

- Traffic Analysis
- Geographical Filtering
- Rate Limiting and Throttling
- Real-time Traffic Analysis and Response

In sum, dynamic proxies can be a useful tool for preventing disruptions to critical infrastructure during DDoS attacks and reducing their impact.

6.0 Conclusion

Overall, dynamic proxies can be a great way to protect SCADA systems from DDoS attacks. But it's important to remember that proxies aren't the only way to stay safe. They should be used as part of a larger security plan. The architecture successfully implements the location-hiding property. For security reasons, no one from outside the destination network can communicate with hosts inside the destination network directly. Data proxies are kept secret unless requested by an authorized user, but setup proxies are made available to anyone who needs them. Without first learning the data proxy used by a specific client, an adversary cannot disrupt ongoing communications between legitimate clients in the network. The use of IP anycast in the architecture aids in tracking down the source of any malicious activity. A sizable botnet can be assembled by the attacker due to the convenience with which the setup proxies can be accessed. The setup proxies allow any part of the architecture to request permission from the target system. If an attacker is within three or four hops of the SPs, the destination can disable the hidden path to stop the attack. Simulations conducted on a network topology at the Internet scale show that a sparse SP density of 0.5% is sufficient to suppress over 93% of attackers even when only 10% of the network is occupied, demonstrating the efficacy of the architecture. The endpoint can determine its proxy assignment procedure and recipient lists thanks to the design.

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